

Kalman filter - for nonlinear systems

Sensors and Systems

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Outline

EKF and UKF

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State
estimation
problem for
nonlinear (NL)
systems

Extended
Kalman filter
(EKF) for the
NL DD
problem

The unscented
transform (UT)

Unscented
Kalman filter
(UKF) for the
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problem

State estimation problem for nonlinear (NL) systems

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State estimation problem for nonlinear (NL) systems

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- ▶ Many real systems are more or less nonlinear.
- ▶ The important questions then are:
 - ▶ Is it possible to estimate states in NL systems?
 - ▶ Does *optimal* estimators exist?
 - ▶ Does *good* estimators exist?

The discrete-discrete (DD) NL estimation problem

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Assume the system:

$$\begin{aligned} x_{k+1} &= f(x_k, u_k) + w_k, \quad w_k \in \text{NID}(\underline{0}, Q_k), \\ y_k &= h(x_k, u_k) + v_k, \quad v_k \in \text{NID}(\underline{0}, R_k), \\ E(w(k)v(l)^T) &= \underline{0} \end{aligned}$$

Given measurements and initial values:

$$\begin{aligned} \mathcal{Y}_k &\triangleq y_0, y_1, y_2 \dots y_k, \quad u_0, u_1, u_2 \dots u_k \\ x_0 &\in N(\hat{x}_0, P_0) \end{aligned}$$

Find: $\hat{x}_k$ which minimized the mean square error

$$E((x_k - \hat{x}_k)^T M (x_k - \hat{x}_k)), \quad M \geq 0$$

The continuous-discrete (CD) NL estimation problem

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The state model is in continuous time i.e. a stochastic differential equation (SDE):

$$dx(t) = f(x(t), u(t))dt + d\omega(t) ,$$

$$\omega(t) \in W(Q(t)) ,$$

$$y(t_k) = h(x(t_k), u(t_k)) + v_k , v_k \in \text{NID}(\underline{0}, R_k)$$

Given measurements and initial values:

$$\mathcal{Y}_k \triangleq y(t_0), y(t_1), y(t_2) \dots y(t_k) , u(s), t_0 \leq s \leq t_k$$

$$x(t_0) \in N(\hat{x}(t_0), P(t_0))$$

Find: $\hat{x}(t)$ which minimized the mean square error

$$E((x(t) - \hat{x}(t))^T M (x(t) - \hat{x}(t))) , M \geq 0$$



Extended Kalman filter (EKF)

Basic idea

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- ▶ The optimal estimator $E(x_k|\mathcal{Y}_k)$ can not normally be obtained.
- ▶ The KF can be used for a linearization of the NL system.
- ▶ This linearization can be made in different ways with fixed or variable linearization point.
- ▶ The best EKF is obtained by using the most resent state estimate as linearization point and using the non linear state and output functions where possible.

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Linearization of f and h with respect to x is needed.

$$F_k(x) \triangleq \frac{\partial f(x)}{\partial x^T} ,$$

$$H_k(x) \triangleq \frac{\partial h(x)}{\partial x^T} .$$

- Notice $F_k \in R^{n \times n}$ and $H_k \in R^{p \times n}$ are matrices.
- The argument x to F_k and H_k should be the most recent estimate of x e.g. $\hat{x}_{k|k-1}$ or $\hat{x}_{k|k}$.

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$$\hat{y}_{k|k-1} = H_k \hat{x}_{k|k-1} + D_k u_k ,$$

$$\tilde{y}_{k|k-1} = y_k - \hat{y}_{k|k-1} ,$$

$$K_k \triangleq P_{k|k-1} H_k^T (H_k P_{k|k-1} H_k^T + R_k)^{-1} ,$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \tilde{y}_{k|k-1} ,$$

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} (I - K_k H_k)^T + K_k R_k K_k^T .$$

Time update from k to $k + 1$:

$$\hat{x}_{k+1|k} = F_k \hat{x}_{k|k} + \Gamma_k u_k ,$$

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$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \tilde{y}_{k|k-1} ,$$

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Time update from k to $k + 1$:

$$\hat{x}_{k+1|k} = f(\hat{x}_{k|k}, u_k) ,$$

$$P_{k+1|k} = F_k P_{k|k} F_k^T + Q_k , \quad F_k = F(\hat{x}_{k|k}) .$$

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Block diagram interpretation - mean value part (no direct feed though)

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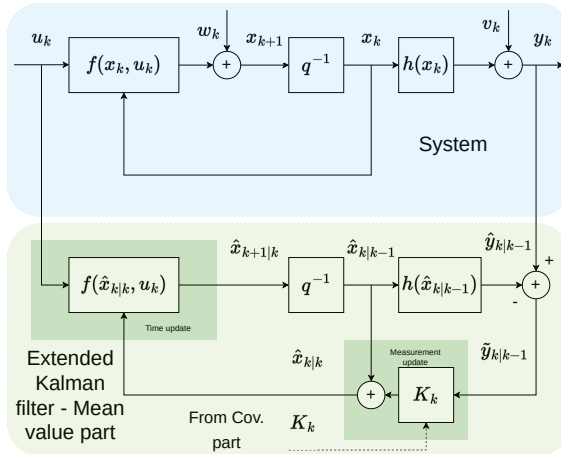
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Block diagram interpretation - covariance value part

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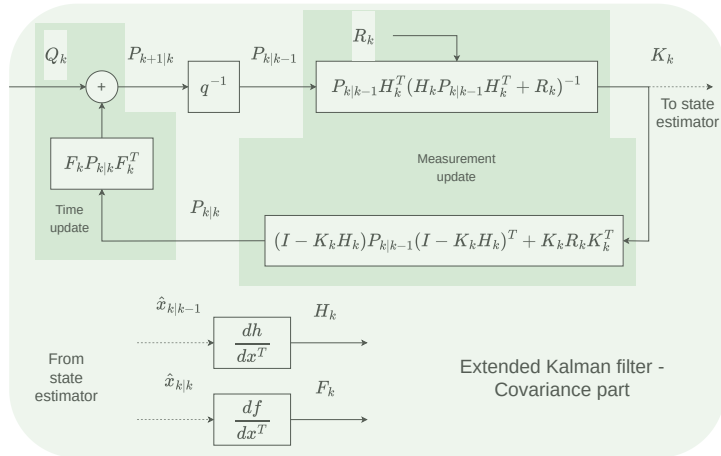
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Important difficulties with the EKF

- ▶ The EKF is not optimal!
- ▶ The EKF is not guaranteed to be stable!

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The unscented transform (UT)

Motivation

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- ▶ In the derivation of the KF for linear systems the linear properties are extensively used and the final algorithm includes the system matrices.
- ▶ Linearization can be applied in the steps before linear system properties are used.

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Example: time update from $\hat{x}_{k|k}, P_{k|k}$ to $\hat{x}_{k+1|k}, P_{k+1|k}$

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Non linear versus linear

In the linear case we have:

$$x_{k+1} = f(x_k, u_k) + w_k = \Phi_k x_k + \Gamma_k u_k + w_k \Rightarrow$$

$$\hat{x}_{k+1|k} = \mathbf{E}(x_{k+1} | \mathcal{Y}_k) \quad (\text{NL})$$

$$= \mathbf{E}(f(x_k, u_k) + w_k | \mathcal{Y}_k) \quad (\text{NL})$$

$$= \mathbf{E}(f(x_k, u_k) | \mathcal{Y}_k) \quad (\text{NL})$$

$$= \mathbf{E}(\Phi_k x_k + \Gamma_k u_k | \mathcal{Y}_k) \quad (\text{L})$$

$$= \Phi_k \hat{x}_{k|k} + \Gamma_k u_k \quad (\text{L})$$

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and

$$P_{k+1|k} = \text{Cov}(x_{k+1}|\mathcal{Y}_k) \quad (\text{NL})$$

$$= \text{Cov}(f(x_k, u_k) + w_k|\mathcal{Y}_k) \quad (\text{NL})$$

$$= \text{Cov}(f(x_k, u_k)|\mathcal{Y}_k) + Q_k \quad (\text{NL})$$

$$= \text{Cov}(\Phi_k x_k + \Gamma_k u_k|\mathcal{Y}_k) + Q_k \quad (\text{L})$$

$$= \Phi_k P_{k|k} \Phi_k + Q_k \quad (\text{L})$$

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Notice also:

$$\begin{aligned} P_{k+1|k} &= \text{Cov}(x_{k+1}|\mathcal{Y}_k) \\ &= \text{E}((x_{k+1} - \text{E}(x_{k+1}|\mathcal{Y}_k))(x_{k+1} - \text{E}(x_{k+1}|\mathcal{Y}_k))^T|\mathcal{Y}_k) \\ &= \text{E}((x_{k+1} - \hat{x}_{k+1|k})(x_{k+1} - \hat{x}_{k+1|k})^T|\mathcal{Y}_k) \\ &= \text{E}(\tilde{x}_{k+1|k}\tilde{x}_{k+1|k}^T) \end{aligned}$$

$$\begin{aligned} \text{Cov}(x_k, y_k|\mathcal{Y}_{k-1}) &= \text{E}((x_k - \text{E}(x_k|\mathcal{Y}_{k-1}))(y_k - \text{E}(y_k|\mathcal{Y}_{k-1}))^T|\mathcal{Y}_{k-1}) \\ &= \text{E}((x_k - \hat{x}_{k|k-1})(y_k - \hat{y}_{k|k-1})^T|\mathcal{Y}_{k-1}) \\ &= \text{E}(\tilde{x}_{k|k-1}\tilde{y}_{k|k-1}^T) \end{aligned}$$

$$\begin{aligned} \text{Cov}(y_k|\mathcal{Y}_{k-1}) &= \text{E}((y_k - \text{E}(y_k|\mathcal{Y}_{k-1}))(y_k - \text{E}(y_k|\mathcal{Y}_{k-1}))^T|\mathcal{Y}_{k-1}) \\ &= \text{E}(\tilde{y}_{k|k-1}\tilde{y}_{k|k-1}^T) \end{aligned}$$

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Reverse derivation

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$$K_k \triangleq \text{Cov}(\tilde{x}_{k|k-1}, \tilde{y}_{k|k-1}) \text{Cov}(\tilde{y}_{k|k-1})^{-1} ,$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \tilde{y}_{k|k-1} ,$$

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$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \tilde{y}_{k|k-1} ,$$

$$P_{k|k} = P_{k|k-1} - K_k \text{Cov}(\tilde{y}_{k|k-1}, \tilde{x}_{k|k-1}) .$$

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$$\hat{y}_{k|k-1} = E(h(x_k, u_k) + v_k | \mathcal{Y}_{k-1}) ,$$

$$\tilde{y}_{k|k-1} = y_k - \hat{y}_{k|k-1} ,$$

$$K_k \triangleq \text{Cov}(x_k, y_k | \mathcal{Y}_{k-1}) \text{Cov}(y_k | \mathcal{Y}_{k-1})^{-1} ,$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \tilde{y}_{k|k-1} ,$$

$$P_{k|k} = P_{k|k-1} - K_k \text{Cov}(y_k, x_k | \mathcal{Y}_{k-1}) .$$

Time update from k to $k+1$:

$$\hat{x}_{k+1|k} = F_k \hat{x}_{k|k} + \Gamma_k u_k ,$$

$$P_{k+1|k} = F_k P_{k|k} F_k^T + Q_k .$$

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Time update from k to $k+1$:

$$\hat{x}_{k+1|k} = E(f(x_k, u_k) + w_k | \mathcal{Y}_k) ,$$

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The generic probabilistic problem

Calculate/approximate the second order statistics of $x, y = f(x)$ given the second order statistics of x and the function f :

$$y = f(x) , \mu_x = E(x) , C_x = \text{Cov}(x) , x \in R^n , y \in R^p$$

For non-linear relations there is no exact solution in general.

The linearization solution

The EKF can be interpreted as using the linearization approach:

$$y = f(\mu_x) + \nabla f(\mu_x)(x - \mu_x) \Rightarrow$$

$$\mu_y = f(\mu_x) , C_{yx} = \nabla f(\mu_x)C_x ,$$

$$C_y = \nabla f(\mu_x)C_x \nabla f(\mu_x)^T .$$

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- ▶ A general approach to approximation in probability problems is *Monte Carlo (MC) simulations*.
- ▶ In MC a number of random samples of x is generated, corresponding $y = f(x)$ are calculated and the empirical estimates of means and covariances are used as approximations.

$$y_i = f(x_i), \quad i = 1, \dots, N, \quad \hat{\mu}_y = \bar{y} = \frac{1}{N} \sum_{i=1}^N f(x_i),$$

$$\hat{C}_{yx} = \frac{1}{N} \sum_{i=1}^N (f(x_i) - \hat{\mu}_y)(x_i - \bar{x})^T, \quad \hat{C}_y = \frac{1}{N} \sum_{i=1}^N (f(x_i) - \hat{\mu}_y)(f(x_i) - \hat{\mu}_y)^T$$

- ▶ The UT method is similar to MC except the realizations of the random variable are not “random” but constructed and the number used is very limited compared to MC.

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$\alpha = 1$, $\kappa = 2$, $\beta = 0$, (Default parameters gives $\lambda = 2$)

$\lambda = \alpha^2(n + \kappa) - n$, $k = \sqrt{n + \lambda}$, $n = \dim(x)$,

$u_i \triangleq$ eigenvector i for C_x , $l_i \triangleq$ eigenvalue i for C_x ,

$$x_i = \begin{cases} \mu_x & , i = 0 \\ \mu_x + k u_i \sqrt{l_i} & , 1 \leq i \leq n \\ \mu_x - k u_i \sqrt{l_i} & , n + 1 \leq i \leq 2n \end{cases}$$

$$w_i^t = \begin{cases} \frac{\lambda}{n + \lambda} & , i = 0 , t = m \text{ (for mean)} \\ \frac{\lambda}{n + \lambda} + 1 - \alpha^2 + \beta & , i = 0 , t = c \text{ (for covarians)} \\ \frac{1}{2(n + \lambda)} & , 1 \leq i \leq 2n \end{cases}$$

$$\hat{\mu}_y = \sum_{i=0}^{2n} w_i^m f(x_i) , \hat{C}_{y_x} = \sum_{i=0}^{2n} w_i^c (f(x_i) - \hat{\mu}_y)(x_i - \hat{\mu}_x)^T ,$$

$$\hat{C}_y = \sum_{i=0}^{2n} w_i^c (f(x_i) - \hat{\mu}_y)(f(x_i) - \hat{\mu}_y)^T .$$

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Important knowledge about the UT

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- ▶ There are many alternative ways to do this. They are some times called *sampling* methods.
- ▶ The UT gives exact results if f is affine $f(x) = Ax + b$ which it also should.
- ▶ For NL f it is still an approximation but often gives good result.
- ▶ Especially it is generally superior to the linearization method and the MC method.

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Example

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In order to have a test where the correct result is known $y = x^T x$ is used with x Gaussian and $E(x) = [1; 1]$ and $Cov(x) = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ (using Matlab notation).

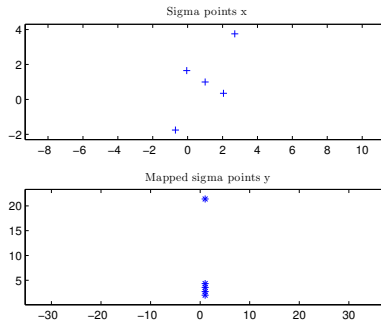


Figure: Sigma points and images for the UT.

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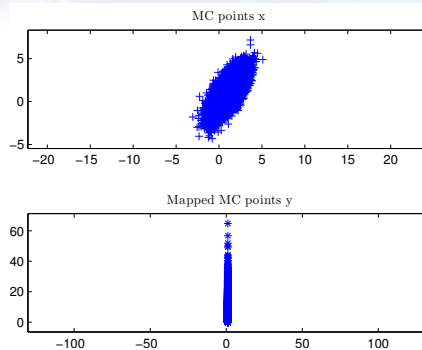


Figure: Random (sigma) points and images for the MC based method.

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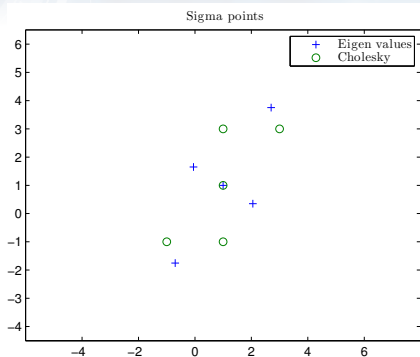


Figure: Comparison of sigma points based on eigenvalues and the Cholesky method.

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Method	My	Cy	Cyx	
UTSCEst	5.000e+0	3.900e+1	4.000e+0	6.000e+0
UTJMEst	5.000e+0	3.100e+1	4.000e+0	6.000e+0
MCEst	5.038e+0	3.385e+1	3.988e+0	6.028e+0
THTK	5.000e+0	3.400e+1	4.000e+0	6.000e+0
UTSCRelE	0	1.471e-1	0	0
UTJMRelE	0	-8.824e-2	0	0
MCRelE	7.663e-3	-4.284e-3	-3.090e-3	4.670e-3

Table: Performance for the above method, the Cholesky based, the MC and the theoretical. The rows marked RelE are the errors relative to the correct results.

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Unscented Kalman filter

Definition

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Assume the NL map f with input x and output y , the UT can then be represented by the function U below:

$$\begin{bmatrix} E(y) \\ \text{Cov}(y, x) \\ \text{Cov}(y) \end{bmatrix} = U(f, E(x), \text{Cov}(x))$$

If the mean and covariance input is conditioned on say z then the output is also conditioned on z i.e.:

$$\begin{bmatrix} E(y|z) \\ \text{Cov}(y, x|z) \\ \text{Cov}(y|z) \end{bmatrix} = U(f, E(x|z), \text{Cov}(x|z))$$

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Initial conditions:

$$x_0 \in N(\hat{x}_{0|-1}, P_{0|-1})$$

Measurements update after receiving y_k and u_k :

$$\begin{bmatrix} \hat{y}_{k|k-1} \\ \text{Cov}(y_k, x_k | \mathcal{Y}_{k-1}) \\ C_k^h \end{bmatrix} = U(h, \hat{x}_{k|k-1}, P_{k|k-1}) ,$$

$$\text{Cov}(y_k | \mathcal{Y}_{k-1}) = C_k^h + R_k ,$$

$$\text{Cov}(x_k, y_k | \mathcal{Y}_{k-1}) = \text{Cov}(y_k, x_k | \mathcal{Y}_{k-1})^T ,$$

$$\tilde{y}_{k|k-1} = y_k - \hat{y}_{k|k-1} ,$$

$$K_k = \text{Cov}(x_k, y_k | \mathcal{Y}_{k-1}) \text{Cov}(y_k | \mathcal{Y}_{k-1})^{-1}$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \tilde{y}_{k|k-1} ,$$

$$P_{k|k} = P_{k|k-1} - K_k \text{Cov}(y_k, x_k | \mathcal{Y}_{k-1}) .$$

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Time update from k to $k + 1$:

$$\begin{bmatrix} \hat{x}_{k+1|k} \\ X \\ C_k^f \end{bmatrix} = U(f, \hat{x}_{k|k}, P_{k|k}) ,$$

$$P_{k+1|k} = \text{Cov}(x_{k+1} | \mathcal{Y}_k) = C_k^f + Q_k .$$

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